



MEMORANDUM

Date: February 3, 2022

To: David Jagger, Energy Solutions; Patrick Kilroy, Navitas Partners; Thomas Mertens, Pacific Gas and Electric Company

From: Evergreen Economics

Re: Air Purifier NPD Analysis Final Draft

This memo details findings from analysis of air purifier sales data, as requested by Pacific Gas and Electric (PG&E) and Energy Solutions. NPD Group provided sales data for the purpose of better understanding the market conditions of key air purifier characteristics.

Research Objectives

The objective of this research is to provide analysis of key air purifier characteristics. Analysis from this research may be used to inform Department of Energy (DOE) appliance test procedures and/or inform ENERGY STAR specifications.

Data Sources

To meet the research objectives, Evergreen Economics has been given access to sales data from NPD Group, which is a market research firm specializing in the acquisition and distribution of sales data for a variety of retail products. NPD Group works with retailers to collect their sales data in exchange for data access and confidentiality guarantees. Data from NPD Group track total sales volume and month sold, in addition to other key characteristics.

NPD Group Data

NPD Group's air purifier data include more than 17 characteristics, including:¹

- Style (floor model, tower, desktop, etc.);

¹ A few variables contained in the NPD data were not included in this analysis as they did not have much variation in the values. Specifically, 97 percent of the units had "No Heater Functionality," 99 percent of the units had "No Humidifier Functionality," and 93 percent of the units had HEPA filtration.

- Ionizer; and
- Price.²

For this analysis, NPD Group has also provided details on the geographic location of sales—whether in California or the rest of the country—and month and year of unit sales. Overall, data from NPD Group contain records for the sale of 7,083,203 air purifiers in the United States between October 2020 and October 2021, 17 percent of which were sold in California. This full dataset was used to select the top 25 model numbers, but for the rest of the analysis presented in this memo, the dataset was restricted to the calendar year from November 2020 to October 2021. Any market share data presented in this memo is the market share of the “Air Purifier-Humidifiers” category and “Air Purifiers” subcategory of products in NPD’s database.

Types of air purifiers reported by NPD include floor models, towers, desktop models, ash tray/plug in models, and personal models. Below are reference photos found online for each air purifier model type, except for ash tray models, for which photos were not available.

Figure 1: Reference Photo for Floor Air Purifier Models



² Price was calculated from NPD sales data of units sold (Dollars/Units).

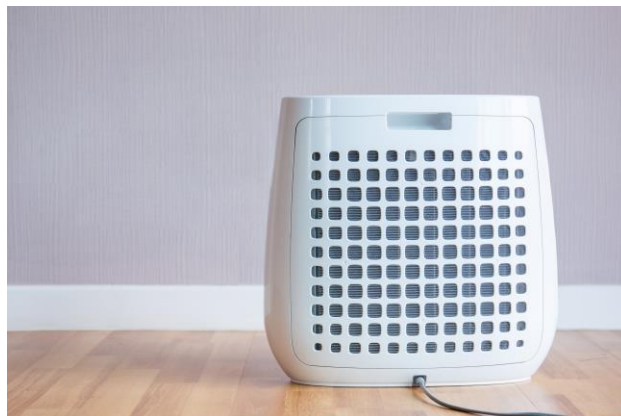
Figure 2: Reference Photo for Tower Style Air Purifier



Figure 3: Reference Photo for Desktop Style Air Purifier



Figure 4: Reference Photo for Personal Air Purifier



Air Purifier Survey Data

Where relevant, this memo draws on findings from a previous survey that Evergreen conducted with a nationally representative sample of air purifier owners in 2021. That survey included questions regarding respondents' air purifier types and brands, motivations for purchasing air purifiers, desirable air purifier features, air purifier usage habits, and spaces cleansed by air purifiers. For more detail about this survey and its methodology, please reference the memo delivered by Evergreen to the PG&E Codes and Standards team on December 15, 2021. These data were collected by Evergreen and did not originate from the data provided by NPD.

Other Data Sources

To supplement the NPD sales data, Evergreen used external data to find information on additional variables requested. We pulled data from the web on the following information:

- Variable fan speeds
- Intelligent controls³
- Marketing terms
- Suggested room size
- ENERGY STAR certification
- California Air Resources Board (CARB) certification⁴
- Clean Air Delivery Rate (CADR) for pollen, dust, and smoke⁵

Where data on these variables are presented, they supplement data provided by NPD and do not originate from NPD. Data relating to variable speed fans, intelligent controls, marketing terms, and suggested room size were gathered from various websites including manufacturers' webpages and retailers such as Lowe's, Home Depot, and Amazon. These variables are referred to broadly as external web data for the remainder of this memo.

The ENERGY STAR specification for room air purifiers measures energy efficiency by using a CADR-to-watt ratio. CADR is a measure of the amount of contaminant-free air delivered by an

³ For this memo, intelligent controls are defined as units that have a sensor that adjusts the fan speed automatically based on air quality conditions.

⁴ California Air Resources Board. "List of CARB-Certified Air Cleaning Devices." <https://ww2.arb.ca.gov/list-carb-certified-air-cleaning-devices>

⁵ Association of Home Appliance Manufacturers. *2021 Directory of Certified Portable Electric Room Air Cleaners – Edition No. 4 – November*. <https://gbs.trutesta.io/aham-indexer-services/aham-indexer/aham/public/AirCleanerFullDir.pdf>

air purifier. To qualify as an ENERGY STAR-certified room air purifier, the unit must meet the key air purifier criteria specified by ENERGY STAR.⁶ The list of ENERGY STAR-certified air purifiers on the energystar.gov website, as of December 13, 2021, contains 317 models.⁷ We flagged models in the analysis either as having the certification or not. Some models in the sales data are discontinued air purifiers and may not be on the current ENERGY STAR list. If information was found regarding ENERGY STAR certification in other data used in this analysis, we flagged those models as well.

Since the enactment of California's Air Cleaner Regulation (AB 2276) in 2010, all room air purifiers sold to people or businesses in California are required to be certified by CARB. We flagged models in the analysis either as having the CARB certification or not.

The Association of Home Appliance Manufacturers (AHAM) is an organization that verifies testing results of home appliances such as air purifiers. Manufacturers voluntarily participate in the AHAM certification program for air purifiers. This certification program uses CADR to determine how well an air cleaner reduces pollutants such as tobacco smoke, dust, and pollen (which represent small, medium, and large sized particles, respectively). We pulled CADR values from the AHAM website, or if unavailable there, from other web sources such as the ENERGY STAR website.⁸

Findings

To inform this analysis, we have focused on key characteristics of air purifiers based on input from PG&E and Energy Solutions—specifically, price segmentation, ENERGY STAR and CARB certification, purchase season, availability of variable fan speeds and intelligent controls, marketing terms, and CADR ratings.

Segmentation

Table 1 presents the segmentation used in the analysis for this memo. The NPD sales data contained model numbers for 62 percent of the units sold between November 2020 and October 2021 in the United States. Of the records with model numbers, Evergreen identified the top 25-selling models in both California and the United States. These top 25 model numbers

⁶ ENERGY STAR. *Room Air Cleaners Key Product Criteria*.

https://www.energystar.gov/products/appliances/air_purifiers_cleaners/key_product_criteria

⁷ <https://www.energystar.gov/productfinder/download/certified-room-air-cleaners/>

⁸ Association of Home Appliance Manufacturers. *2021 Directory of Certified Portable Electric Room Air Cleaners – Edition No. 4 – November*. <https://gbs.trutesta.io/aham-indexer-services/aham-indexer/aham/public/AirCleanerFullDir.pdf>

accounted for 38 percent of the total units sold between November 2020 and October 2021 (and 61 percent of the total units with a model number).⁹ A full list of the top 25 models and manufacturers with number of units sold is available in the Appendix.¹⁰

Table 1: Segmentation Used in Analysis

Segment	Sub-Segment	California			United States		
		Net Units Sold	% Units Sold in CA	Net \$ Sold (in Millions)	Net Units Sold	% Units Sold in US	Net \$ Sold (in Millions)
Top 25	with model #	436,037	41%	\$72.7	2,435,237	38%	\$385.7
All Others	with model #	241,686	23%	\$47.2	1,563,685	24%	\$296.0
All Others	without model #	381,029	36%	\$44.3	2,482,560	38%	\$260.2
Total	Total	1,058,752	100%	\$164.3	6,481,482	100%	\$941.8

As mentioned previously, the NPD sales data contained model numbers for 62 percent of the units sold between November 2020 and October 2021 in the United States. The total number of units sold and the percentage of units with model numbers is presented in Table 2, by segment.

Almost none of the ash tray/plug-in style units had model numbers in the NPD data. The floor model-style purifiers had a lower-than-average percent of units with model numbers (54% compared to 62% of units in the United States overall), and tower, personal, and desktop style purifiers had higher than average percent of units with model numbers (78%, 77%, and 65%, respectively). Also of note is that the lower the price of a unit, the less likely it was to have a model number in the NPD data. The external web data were looked up and merged by model number, so the units that did not have model numbers will not be included in the tables or figures that present the following information: variable fan speeds, intelligent controls, marketing terms, suggested room size, ENERGY STAR certification, CARB certification, and CADR for pollen, dust, and smoke.

⁹ This segmentation into top 25 models and all other models was done by Evergreen on the basis of NPD data. NPD did not segment the data in this manner.

¹⁰ Evergreen ranked models based on number of units sold between October 2020 to October 2021. "Top models" were the top 25 model numbers in terms of units sold.

Table 2: Percentage of Units in the NPD Data with Model Number

Feature	Segment	Percentage of Units with Model Number	Total Number of Units
All Models	California	64%	1,058,752
	United States	62%	6,481,482
Price	< \$75	36%	1,395,042
	\$75 - \$99	59%	1,668,857
	\$100 - \$199	63%	1,765,379
	\$200 +	84%	1,652,204
Season ¹¹	Winter	64%	1,615,608
	Spring	59%	1,578,028
	Summer	60%	1,694,523
	Fall	65%	1,593,323
Style	Floor Model	54%	3,045,977
	Tower	78%	1,884,900
	Desktop	65%	1,294,006
	Ash Tray/Plug-In	0%	202,159
	Personal	77%	54,422

¹¹ Season was categorized from the month provided in the NPD sales data (Winter = Dec, Jan, Feb; Spring = Mar, Apr, May; Summer = Jun, Jul, Aug; Fall = Sep, Oct, Nov).

Characteristics by Geography and Top Selling Models

Table 3 presents the information that was provided in the NPD sales data—specifically, price, season sold, ionizer, and style for the top models and for all models sold in California and the entire United States. Strong seasonality trends were not observed, with roughly equal shares of models sold across all four seasons, both for the top selling models and for all models sold in California and the entire United States. Top models tended to be slightly more expensive than all models overall, with top selling models 6 percent more likely to be priced at \$200 or more compared to all models sold in the United States. These numbers are also visualized in Figure 5. Average prices were higher for top selling models, with top selling models in California \$6 more expensive on average, and top selling models across the United States \$13 more expensive on average, when compared to all models sold in those regions.

Purifiers with ionizers proved to be uncommon, though they accounted for a slightly larger share of units sold across the United States (23%) compared to units sold in California (19%). The distribution of the style of the purifier was noticeably different when comparing the top-selling models to all models sold. Tower-style purifiers made up the largest percentage of top selling models (over 42%); however, the floor model-style purifiers made up the largest percentage of all models sold (over 47%). Despite the apparent popularity of tower models, floor models accounted for a larger share of all models, both across the United States (47%) and in California (51%). These numbers are also visualized in Figure 6.

Table 3: Characteristics of Air Purifiers Sold Between November 2020 and October 2021 – NPD Data

Feature	Segment	Percentage of Units			
		Top Models California (N=436,037)	Top Models U.S. (N=2,435,237)	All Models California (N=1,058,752)	All Models U.S. (N=6,481,482)
Price	Average	\$167	\$158	\$155	\$145
	< \$75	10%	10%	16%	22%
	\$75 - \$99	25%	32%	25%	26%
	\$100 - \$199	32%	27%	32%	27%
	\$200 +	32%	31%	27%	25%
Season	Winter	25%	26%	24%	25%
	Spring	23%	23%	24%	24%
	Summer	26%	25%	28%	26%
	Fall	27%	26%	25%	25%
Ionizer	No Ionizer	80%	82%	81%	77%
	Ionizer	20%	18%	19%	23%
Style	Floor Model	35%	34%	51%	47%
	Tower	42%	43%	29%	29%
	Desktop	22%	23%	18%	20%
	Ash Tray/Plug-In	0%	0%	2%	3%
	Personal	1%	0%	1%	1%

To better visualize the price and style data, Figure 5 displays the percentage of units sold by price and geography, and Figure 6 displays the percentage of units sold by style and geography.

Figure 5: Percentage of Units Sold by Price (Nov 2020 - Oct 2021)

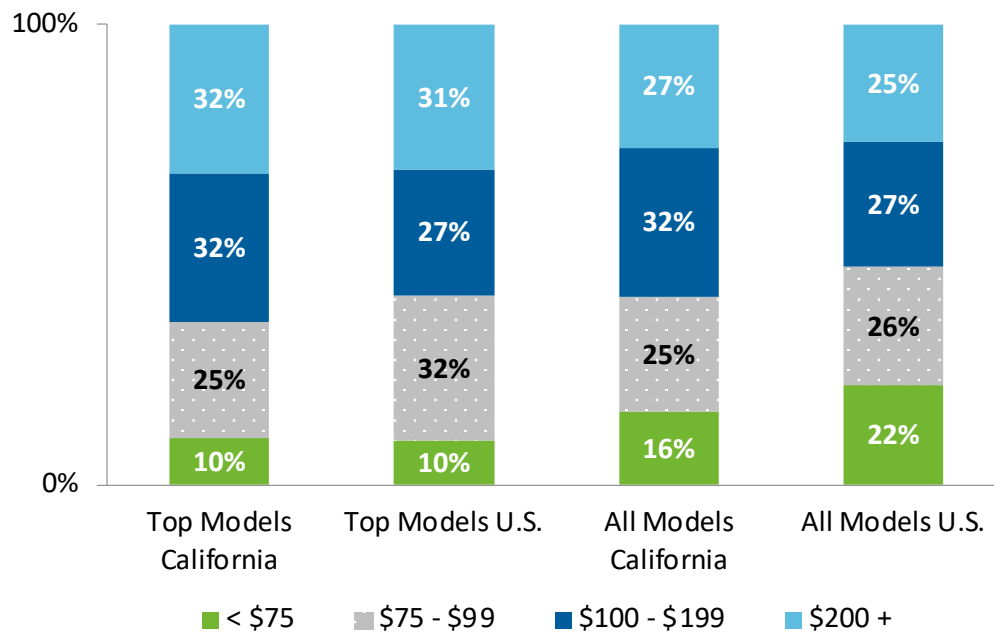
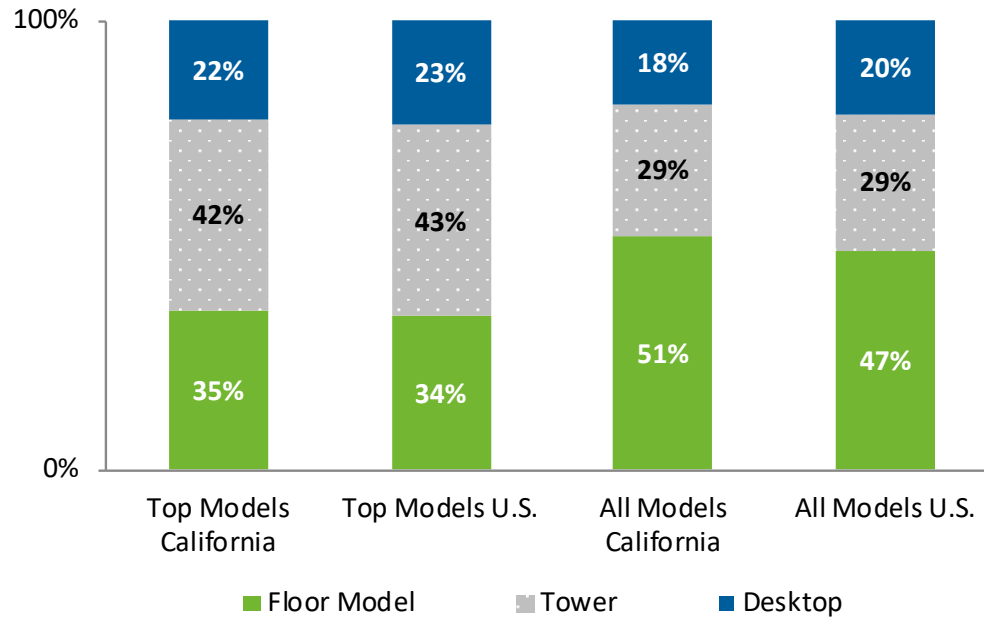


Figure 6: Percentage of Units Sold by Style (Nov 2020 - Oct 2021)



Evergreen also pulled in information from other data sources and merged that information with NPD sales data by model number.¹² Table 4 presents these additional data for the top models sold, as well as all models sold in California and the United States. Sixty-two percent of the units sold had a model number in the NPD data. The 38 percent of units without a model number are not represented in the table below. Merged data included the availability of variable speed fans and intelligent controls, and the marketing terms.¹³

To systematically determine the marketing terms, we analyzed all text from the product sales page and assigned the variable if it was mentioned. For example, if the product sales page mentioned “allergies” and “COVID-19,” they would be considered as marketing for allergies *and* COVID-19. Table 4 displays the findings for the top models sold, as well as all models sold in California and across the United States.

Nearly all models reviewed had variable fan speeds, while intelligent controls are far less common. Units sold in California had intelligent controls more often (41%) when compared to

¹² These other data sources are described under the “Other Data Sources” section found on page 2. Data sources described therein did not originate from the data provided by NPD.

¹³ For this memo, a unit with intelligent controls is defined as a unit that has a sensor that adjusts the fan speed automatically based on air quality conditions.

units sold across the United States (35%); a similar trend was seen in the top-selling models (43% in California versus 33% in the United States). The low incidence of intelligent controls coheres with self-reports from our air purifier survey, where 29 percent of respondents indicated that their purifier could adjust its settings automatically.

The distribution of marketed features was similar for both top models and all models, within California and across the United States. Allergies were the most common advertised feature accounting for nearly all reviewed purifiers, which aligns with self-reported purchasing motivations from our air purifier survey indicating that reducing allergens were the most important motivations for purchasing an air purifier.¹⁴ Pollen and mold particles are on the larger end of the particle spectrum, followed by dust, dander, bacteria, smoke, and viruses. In general, the marketing terms used, correlate to the size of the particle (i.e., the larger the particle, the more often the marketing term is used, with the exception of mold).

Reviewed purifiers tended to purify small- or medium-sized spaces, with 37 percent of units sold in California and 38 percent of units sold across the United States purifying spaces up to 199 square feet, while another 29 percent of units sold in California and 25 percent of units sold in the United States purified a maximum area of 300 to 399 square feet. Though fewer purifiers sold between November 2020 and October 2021 were for medium-sized rooms than the purifiers reported in the air purifier survey, the overall distribution was similar, with the bulk of purifiers cleansing spaces of less than 400 square feet.

¹⁴ These data were collected by Evergreen for a previous memo as discussed in the preceding section regarding Air Purifier Survey Data. This information did not originate from data provided by NPD.

Table 4: Characteristics of Air Purifiers Sold Between November 2020 and October 2021 – Web Data

Feature ¹⁵	Segment	Percentage of Units*			
		Top Models California (N=436,037)	Top Models U.S. (N=2,435,237)	All Models California (N=677,723)	All Models U.S. (N=3,998,922)
Certifications	ENERGY STAR	65%	67%	66%	67%
	CARB	100%	100%	100%	100%
Fan / Controls	Variable Speed Fan	100%	100%	100%	100%
	Intelligent Controls	43%	33%	41%	35%
Marketing Terms	Allergies	100%	100%	99%	99%
	Odor	98%	98%	93%	92%
	Pet Dander	85%	88%	83%	84%
	Smoke	69%	70%	74%	76%
	Bacteria	47%	46%	43%	41%
	Mold	31%	34%	36%	38%
	Pollutants	35%	28%	35%	33%
	Viruses	27%	28%	27%	28%
	COVID-19	3%	3%	5%	5%
Marketed Room Size (Square Feet)	<i>Average</i>	<i>293 sq ft</i>	<i>283 sq ft</i>	<i>305 sq ft</i>	<i>300 sq ft</i>
	< 150 Sq Ft	5%	5%	9%	9%
	150 - 199 Sq Ft	31%	35%	28%	29%
	200 - 299 Sq Ft	0%	4%	5%	7%
	300 - 349 Sq Ft	15%	13%	12%	10%
	350 - 399 Sq Ft	20%	15%	17%	15%
	400 - 499 Sq Ft	4%	4%	4%	5%
	500 + Sq Ft	11%	12%	13%	13%

¹⁵ None of the “Features” presented in this table originated from data provided by NPD.

Feature ¹⁵	Segment	Percentage of Units*			
		Top Models California (N=436,037)	Top Models U.S. (N=2,435,237)	All Models California (N=677,723)	All Models U.S. (N=3,998,922)
	Not Specified	15%	12%	13%	12%

*Features in this table were merged by model number. 62 percent of the units sold had a model number; the 38 percent without a model number are not represented in this table.

Table 5 presents CADR segments for the top models sold, as well as all models sold in California and across the United States. CADR information was gathered from the AHAM website, the ENERGY STAR website, or from the product information found on the web, and did not originate from data provided by NPD. The CADR values were merged with the NPD sales data by model number. Sixty-two percent of the units sold had a model number; however, we were unable to find CADR information for all model numbers. The units represented in Table 5 account for 41 percent of the total units sold between November 2020 and October 2021. The distribution of CADRs was similar for pollen and dust, and generally lower for smoke. The average CADR decreased as the particle size decreased.

Table 5: Characteristics of Air Purifiers Sold Between November 2020 and October 2021 – Web Data

Feature ¹⁶	Segment	Percentage of Units*			
		Top Models California (N=281,266)	Top Models U.S. (N=1,633,747)	All Models California (N=449,969)	All Models U.S. (N=2,678,146)
CADR Pollen (Large Particle)	<i>Average</i>	<i>197 CADR</i>	<i>182 CADR</i>	<i>198 CADR</i>	<i>193 CADR</i>
	< 50	0%	0%	0%	0%
	50 - 99	5%	8%	10%	12%
	100 - 149	36%	46%	34%	37%
	150 - 199	3%	3%	5%	6%
	200 - 249	32%	22%	25%	19%
	250 - 299	11%	8%	10%	10%

¹⁶ None of the “Features” presented in this table originated from data provided by NPD.

Feature ¹⁶	Segment	Percentage of Units*			
		Top Models California (N=281,266)	Top Models U.S. (N=1,633,747)	All Models California (N=449,969)	All Models U.S. (N=2,678,146)
	300 - 349	6%	7%	5%	6%
	350 +	7%	7%	10%	10%
CADR Dust (Medium Particle)	<i>Average</i>	<i>194 CADR</i>	<i>181 CADR</i>	<i>192 CADR</i>	<i>187 CADR</i>
	< 50	0%	0%	0%	0%
	50 - 99	5%	8%	10%	11%
	100 - 149	36%	46%	35%	38%
	150 - 199	3%	3%	5%	6%
	200 - 249	43%	30%	31%	25%
	250 - 299	0%	0%	4%	4%
	300 - 349	6%	7%	7%	7%
	350 +	7%	7%	8%	8%
CADR Smoke (Small Particle)	<i>Average</i>	<i>187 CADR</i>	<i>174 CADR</i>	<i>185 CADR</i>	<i>180 CADR</i>
	< 50	3%	3%	3%	3%
	50 - 99	28%	32%	26%	28%
	100 - 149	11%	13%	15%	16%
	150 - 199	0%	6%	5%	8%
	200 - 249	45%	33%	32%	26%
	250 - 299	0%	0%	5%	5%
	300 - 349	6%	7%	6%	6%
	350 +	7%	7%	8%	8%

*Features in this table were merged by model number. Sixty-two percent of the units sold had a model number; however, we were unable to find CADR information for all model numbers. The units represented in this table account for 41 percent of the total units sold between November 2020 and October 2021.

Characteristics by Season

This next section characterizes the market transformation and growth rate of air purifiers by season. Table 6 presents the air purifier features from the NPD data by season. The average price of units sold in the summer and fall (\$152) were higher than the average price of units sold in the winter and spring (\$138-\$139). This price difference may have been a result of the distribution of styles purchased in each season. More desktop and ash tray/plug-in purifiers were purchased in the winter and spring (27% and 24%), compared to the summer and fall (20% and 23%).

Table 6: Characteristics of Air Purifiers Sold Between November 2020 and October 2021 – NPD Data

Feature	Segment	Percentage of Units			
		Winter (N=1,615,608)	Spring (N=1,578,028)	Summer (N=1,694,523)	Fall (N=1,593,323)
Price	Average	\$139	\$138	\$152	\$152
	< \$75	24%	26%	19%	18%
	\$75 - \$99	27%	23%	24%	28%
	\$100 - \$199	24%	27%	28%	29%
	\$200 +	24%	23%	28%	26%
Ionizer	No Ionizer	76%	77%	78%	76%
	Ionizer	24%	23%	22%	24%
Style	Floor Model	42%	46%	52%	47%
	Tower	30%	29%	28%	30%
	Desktop	23%	19%	18%	20%
	Ash Tray/Plug-In	4%	5%	2%	2%
	Personal	1%	1%	1%	1%

Table 7 presents the air purifier features from the web data by season. The external web data were merged with NPD sales data by model number. Sixty-two percent of the units sold had a

model number.¹⁷ The 38 percent of units without a model number are not represented in the table below. The marketing terms varied somewhat across the seasons, possibly driven by the size and style of the units purchased during each season. Some of the features gathered from the web data (and shown in the table below) are also visualized by month in Figure 9 through Figure 12.

Table 7: Characteristics of Air Purifiers Sold Between November 2020 and October 2021 – Web Data

Feature ¹⁸	Segment	Percentage of Units*			
		Winter (N=1,033,486)	Spring (N=926,545)	Summer (N=1,008,922)	Fall (N=1,029,969)
Certifications	ENERGY STAR	69%	66%	65%	67%
	CARB	100%	100%	100%	100%
Fan / Controls	Variable Speed Fan	99%	100%	100%	100%
	Intelligent Controls	31%	35%	37%	35%
Marketing Terms	Allergies	99%	99%	99%	99%
	Odor	92%	91%	94%	93%
	Pet Dander	83%	82%	85%	86%
	Smoke	78%	72%	74%	78%
	Bacteria	44%	39%	39%	43%
	Mold	40%	35%	36%	41%
	Pollutants	27%	33%	36%	34%
	Viruses	32%	24%	25%	30%
	COVID-19	3%	4%	7%	5%
Marketed Room Size (Square Feet)	<i>Average</i>	<i>290 sq ft</i>	<i>294 sq ft</i>	<i>311 sq ft</i>	<i>302 sq ft</i>
	< 150 Sq Ft	11%	9%	10%	7%
	150 - 199 Sq Ft	28%	30%	27%	32%
	200 - 299 Sq Ft	7%	6%	6%	7%

¹⁷ Web data is described under the “Other Data Sources” section found on page 2. “Features” in Table 7 did not originate from data provided by NPD.

¹⁸ None of the “Features” presented in this table originated from data provided by NPD.

Feature ¹⁸	Segment	Percentage of Units*			
		Winter (N=1,033,486)	Spring (N=926,545)	Summer (N=1,008,922)	Fall (N=1,029,969)
	300 - 349 Sq Ft	11%	9%	10%	10%
	350 - 399 Sq Ft	14%	12%	16%	17%
	400 - 499 Sq Ft	5%	5%	5%	4%
	500 + Sq Ft	13%	13%	15%	13%
	Not Specified	10%	16%	12%	11%

*Features in this table were merged by model number. Sixty-two percent of the units sold had a model number; the 38 percent without a model number are not represented in this table.

To better visualize the seasonal trends, the figures in this section are presented by month and year.

Figure 7 presents the number of air purifiers purchased in the United States between November 2020 and October 2021, by region. No strong seasonal effect was observed, even though purchases peaked at least once in winter, spring, summer, and fall, with two peaks during the summer (June and August) and two peaks in the fall (September and November).

Figure 7: Number of Units Sold per Month

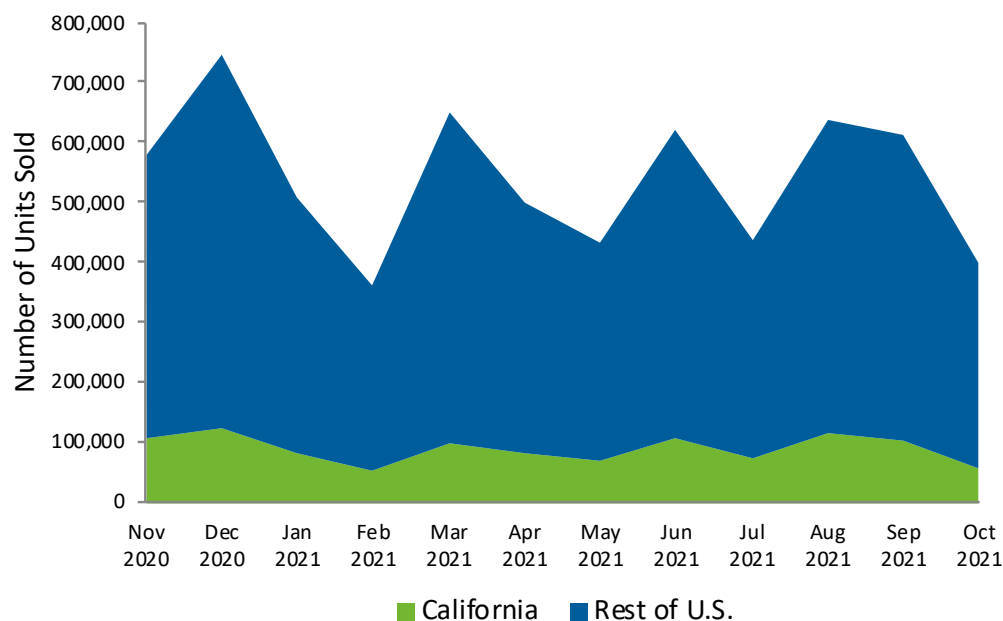
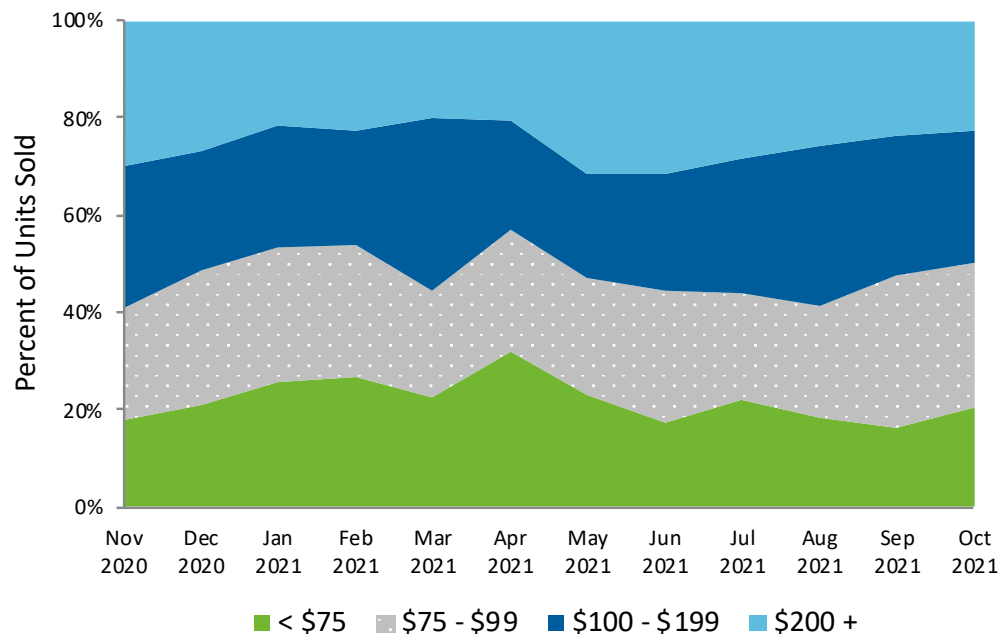


Figure 8 presents the percentage of units sold each month that falls within each of the four price bins segmented by Evergreen. Purifiers at different price points were generally equally distributed over time, with each price bin comprising about 20 percent of the units that were sold in a given month. These trends tended to shift in May and June of 2021, when more expensive purifiers (i.e., purifiers costing more than \$200) gained roughly 10 percent of purifiers sold from April to May of 2021. In general, these trends point to an air purifier market that is not strongly driven by preferences for lower prices, as the bulk of the units sold since November 2020 have been of units of moderate or higher price.

Figure 8: Percentage of Units Sold per Month by Price Bin



The largest share of units sold over all seasons are purifiers marketed for relatively smaller spaces, such as bedrooms (i.e., maximum room size of less than 200 square feet; Figure 9). The marketed room size has been trending slightly upward since November 2020. The room size information presented in Figure 9 was gathered from data found on the web, where available and joined with the sales data provided by NPD.¹⁹

¹⁹ “Marketed Maximum Room Size” was not included in the data provided by NPD.

Figure 9: Percentage of Units Sold per Month by Marketed Maximum Room Size

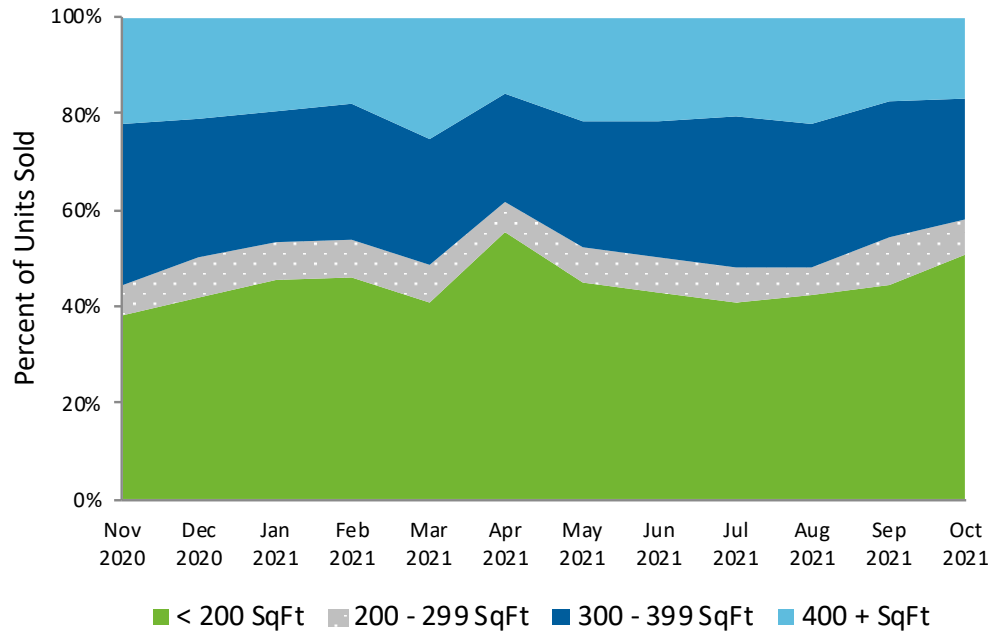


Figure 10 presents the percentage of units sold each month that referenced “smoke” or “pollutants” in the product information found on the web that was joined onto existing NPD data.²⁰ Units sold that were found to have “smoke” in their marketing messaging tended to be stable over the course of the time period, hovering around 80 percent of purifiers sold except for April through July 2021, where there was a 10 to 15 percent drop. This drop coincided with a similar rise in the sale of units that were found to have “pollutant” in their marketing messaging.

²⁰ “Marketing Term” was not included in the data provided by NPD.

Figure 10: Percentage of Units Sold per Month by Marketing Terms

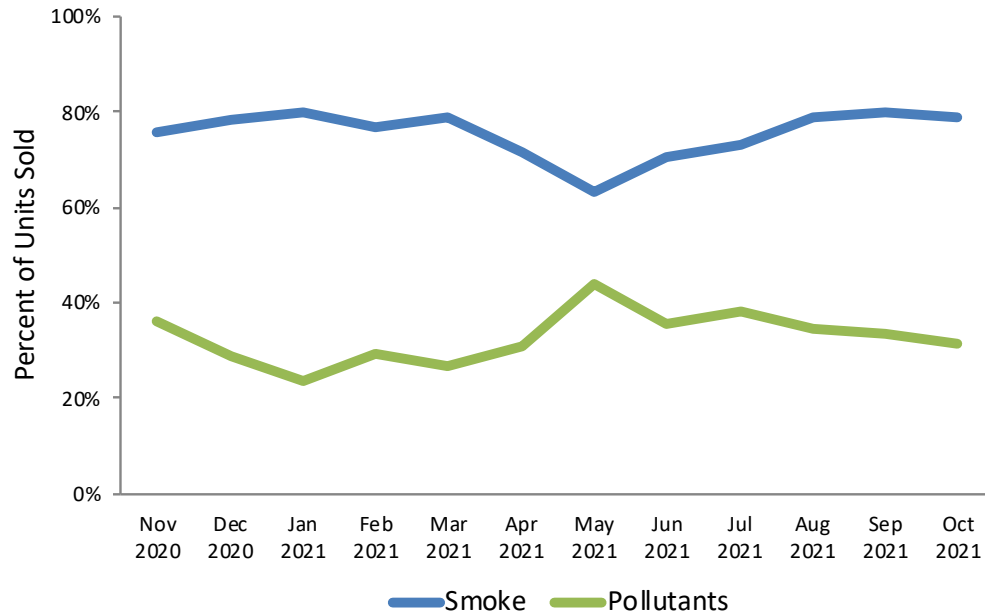


Figure 11 presents the percentage of units sold each month that referenced “bacteria,” “viruses,” or “COVID-19” in the product information found on the web that was joined onto the NPD data. In contrast with marketing messaging oriented around smoke reduction, messaging surrounding the reduction of bacteria and viruses was much less common, and the reduction of COVID-19 especially uncommon, with the share of reviewed purifiers that highlight COVID-19 consistently under 10 percent. Since November 2020, the number of units sold that use “COVID-19” in their marketing messaging has been increasing slightly, while units sold that use “bacteria” in their marketing messaging have been decreasing slightly.

Since the beginning of the pandemic, the Environmental Protection Agency (EPA) has enforced rules governing false or misleading claims regarding viral reduction.²¹ Though mentions of viral reduction are relatively low (ranging from 30 to 40%), they still are far more frequent than mentions of COVID-19, possibly due to the risk of making inaccurate claims related to the pandemic.

²¹ Adam Allington “Air Purifiers Making Covid-19 Claims Face Legal, Financial Risks.” *bloomberglaw.com*. April 23, 2020. <https://news.bloomberglaw.com/environment-and-energy/air-purifiers-making-covid-19-claims-face-legal-financial-risks>

Figure 11: Percentage of Units Sold per Month by Marketing Terms

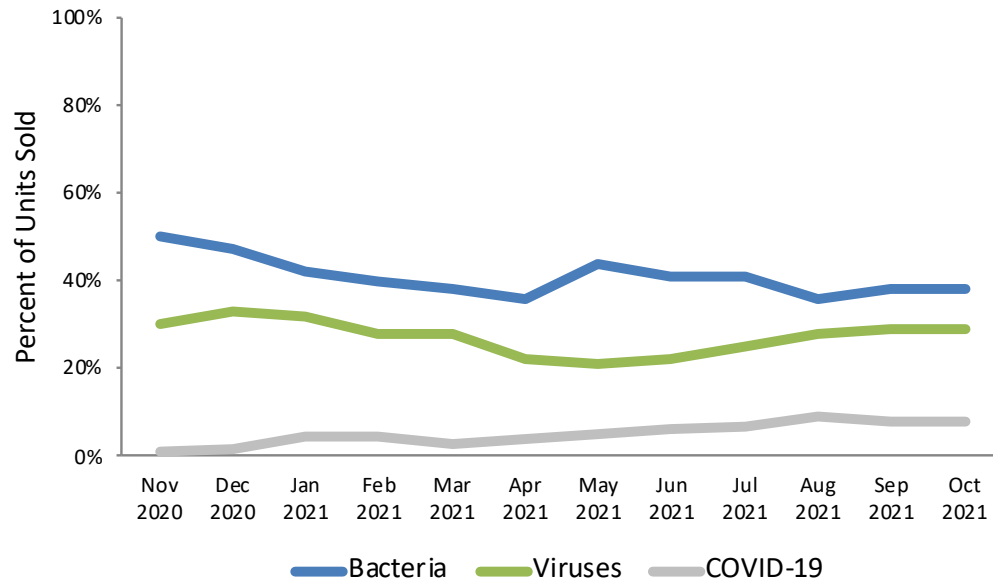
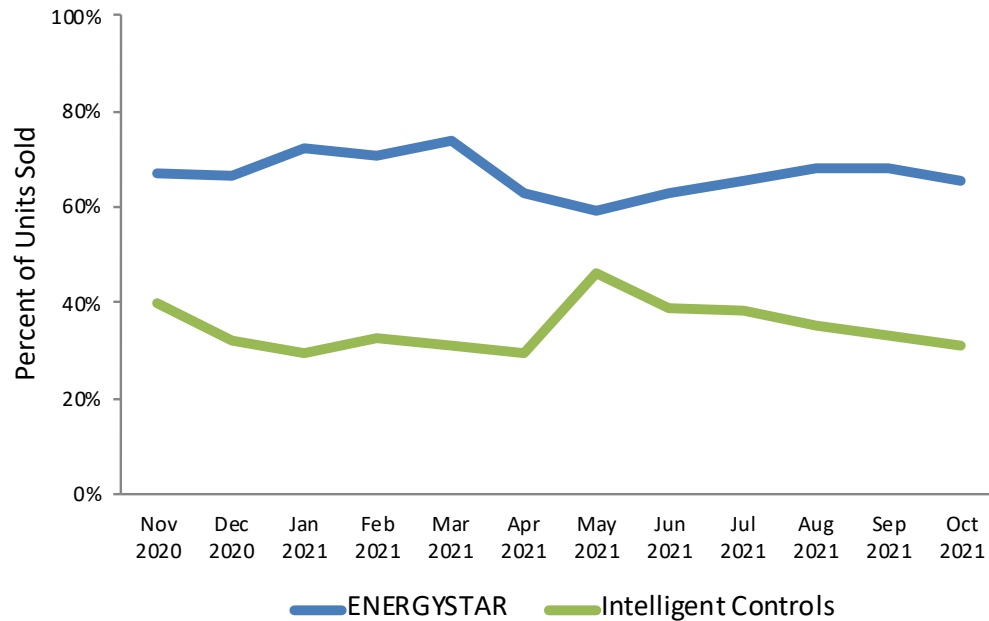


Figure 12 presents the percentage of units sold each month that referenced “intelligent controls” in the product information found on the web or was flagged as ENERGY STAR certified either on the ENERGY STAR website, or from other information found on the web that was joined onto the NPD data. In general, ENERGY STAR units and units with intelligent controls had a relatively flat trend between November 2020 and October 2021. The popularity of units with intelligent controls hovered between 30 and 40 percent of units over time, with the exception of a spike to 46 percent in May 2021. The strong upswing in units purchased with intelligent controls in May 2021 could have been motivated by seasonal increases in allergens.

Figure 12: Percentage of Units Sold per Month by Marketing Terms



Characteristics by Price

This next section visualizes the air purifier features found at four price points segmented by Evergreen. Table 8 presents the air purifier features from the NPD data by price. Ionizers are found most often in the lowest priced units. The desktop style made up a majority of the lowest priced units, and ash tray/plug-in and personal style units were only found in the lowest priced units. Floor model and tower style units made up most of the units in the \$75-\$99 price range. Purifiers that are over \$100 comprised a majority of floor style units.

Table 8: Characteristics of Air Purifiers Sold Between November 2020 and October 2021 – NPD Data

Feature	Segment	Percentage of Units			
		< \$75 (N=1,395,042)	\$75 - \$99 (N=1,668,857)	\$100 - \$199 (N=1,765,379)	\$200 + (N=1,652,204)
Season	Winter	28%	27%	22%	24%
	Spring	29%	22%	24%	22%
	Summer	23%	25%	27%	29%
	Fall	20%	27%	26%	25%
Ionizer	No Ionizer	51%	91%	82%	78%
	Ionizer	49%	9%	18%	21%
Style	Floor Model	12%	43%	69%	58%
	Tower	17%	40%	27%	30%
	Desktop	53%	18%	4%	11%
	Ash Tray/Plug-In	14%	0%	0%	0%
	Personal	4%	0%	0%	0%

Table 9 presents the air purifier features from the web data by price. The web data were merged with NPD sales data by model number. Sixty-two percent of the units sold had a model number. The 38 percent of units without a model number are not represented in the table below. The marketing terms varied somewhat across the price points, possibly driven by the size and style of the units purchased within each bin. ENERGY STAR-certified units were found more often in the units that cost \$75 or higher. Intelligent controls were only found in units above \$100. As expected, average marketed room size increased as price increased. Some of the features gathered from the web data (and shown in the table below) are also visualized in Figure 13 through Figure 15.

Table 9: Characteristics of Air Purifiers Sold Between November 2020 and October 2021 – Web Data

Feature ²²	Segment	Percentage of Units*			
		< \$75 (N=506,956)	\$75 - \$99 (N=985,125)	\$100 - \$199 (N=1,115,867)	\$200 + (N=1,390,974)
Certifications	ENERGY STAR	27%	72%	78%	69%
	CARB	100%	100%	100%	100%
Fan / Controls	Variable Speed Fan	97%	100%	100%	100%
	Intelligent Controls	0%	0%	45%	63%
Marketing Terms	Allergies	96%	100%	100%	99%
	Odor	93%	98%	96%	86%
	Pet Dander	91%	96%	70%	84%
	Smoke	98%	80%	84%	58%
	Bacteria	19%	59%	46%	35%
	Mold	30%	54%	33%	33%
	Pollutants	14%	8%	33%	57%
	Viruses	17%	56%	31%	9%
	COVID-19	0%	0%	8%	7%
Marketed Room Size (Square Feet)	<i>Average</i>	<i>121 sq ft</i>	<i>222 sq ft</i>	<i>275 sq ft</i>	<i>481 sq ft</i>
	< 150 Sq Ft	51%	11%	1%	0%
	150 - 199 Sq Ft	39%	59%	35%	1%
	200 - 299 Sq Ft	0%	11%	6%	6%
	300 - 349 Sq Ft	0%	0%	28%	6%
	350 - 399 Sq Ft	0%	0%	27%	21%
	400 - 499 Sq Ft	0%	0%	0%	13%
	500 + Sq Ft	0%	17%	1%	25%
	Not Specified	10%	2%	3%	27%

*Features in this table were merged by model number. 62 percent of the units sold had a model number; the 38 percent without a model number are not represented in this table.

²² None of the “Features” presented in this table originated from data provided by NPD.

Table 10 presents CADR/Watt by price. CADR/Watt was obtained from the ENERGY STAR website and was only available for ENERGY STAR certified units that are on the current list. This data was found via a web search and did not originate in the NPD data set. It should be noted that there is a minimum smoke CADR/Watt to be certified as ENERGY STAR, so the values below may be higher than the CADR/Watt found in all units sold in the United States.²³ The CADR/Watt values were merged with NPD sales data by model number to weight the values by number of units sold. The units represented in this table account for 37 percent of the total units sold between November 2020 and October 2021. In general, the price increases as the CADR/Watt increases, with the exception being units that have a CADR/Watt between 3.5 and 4.9. The highest average CADR/Watt is found in units in the \$100-\$199 price range (4.7 to 5.2 CADR/Watt).

Table 10: Characteristics of Air Purifiers Sold Between November 2020 and October 2021 – Web Data

Feature ²⁴	Segment	Percentage of Units*					Average Price (N=2,416,829)
		< \$75 (N=88,072)	\$75 - \$99 (N=679,794)	\$100-\$199 (N=825,196)	\$200 + (N=823,767)	Total (N=2,416,829)	
CADR/ Watt Pollen	Average	2.1	2.8	5.2	4.3	4.1	
	< 2.5	91%	66%	5%	0%	5%	\$72
	2.5 - 2.9	9%	25%	28%	4%	46%	\$119
	3.0 - 3.4	0%	6%	11%	45%	8%	\$200
	3.5 - 4.9	0%	0%	38%	16%	9%	\$172
	5 +	0%	4%	18%	35%	33%	\$208
CADR/ Watt Dust	Average	2.1	2.7	5.1	4.2	4.0	
	< 2.5	46%	57%	2%	0%	23%	\$85
	2.5 - 2.9	54%	34%	26%	27%	18%	\$117
	3.0 - 3.4	0%	3%	8%	22%	21%	\$188
	3.5 - 4.9	0%	3%	23%	7%	19%	\$182
	5 +	0%	4%	41%	44%	19%	\$226

²³ ENERGY STAR. *Room Air Cleaners Key Product Criteria*.

https://www.energystar.gov/products/appliances/air_purifiers_cleaners/key_product_criteria

²⁴ None of the "Features" presented in this table originated from data provided by NPD.

Feature ²⁴	Segment	Percentage of Units*					Average Price (N=2,416,829)
		< \$75 (N=88,072)	\$75 - \$99 (N=679,794)	\$100-\$199 (N=825,196)	\$200 + (N=823,767)	Total (N=2,416,829)	
CADR/ Watt Smoke	Average	2.2	2.5	4.7	4.1	3.8	
	< 2.5	68%	9%	0%	0%	18%	\$85
	2.5 - 2.9	28%	84%	33%	29%	30%	\$131
	3.0 - 3.4	1%	3%	2%	18%	11%	\$194
	3.5 - 4.9	0%	0%	21%	6%	11%	\$159
	5 +	4%	4%	44%	47%	30%	\$212

*Features in this table were merged by model number. Sixty-two percent of the units sold had a model number; however, we were unable to find CADR/Watt information for all model numbers. The CADR/Watt information was obtained from the ENERGY STAR website and only represents ENERGY STAR certified units. The units represented in this table account for 37 percent of the total units sold between November 2020 and October 2021.

To better visualize the price data, figures created using the data from the tables above are presented below.

Figure 13 presents the percentage of units sold within each price bin that referenced “smoke” or dander” in the product information found on the web that was joined onto the NPD data. These use of these two marketing terms trends downwards as the price point increases.

Figure 13: Percentage of Units Sold per Price Bin by Marketing Terms

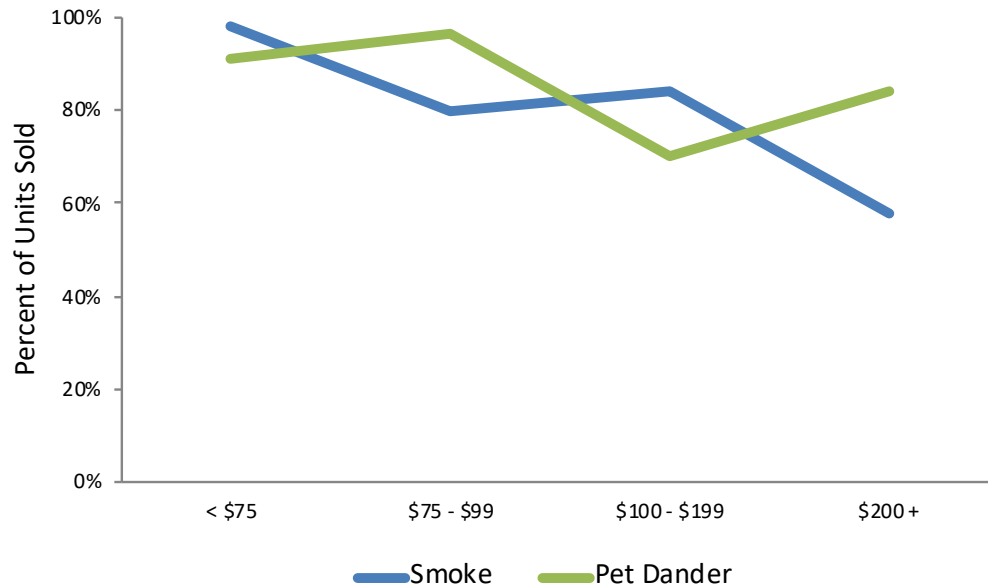


Figure 14 presents the percentage of units sold within each price bin that referenced “intelligent controls,” “pollutants,” or “COVID-19” in the product information found on the web or was flagged as ENERGY STAR-certified either on the ENERGY STAR website, or from other information found on the web that was joined onto the NPD data. The use of these four marketing terms tends to increase as the price point increases. This was especially true of ENERGY STAR models, with only roughly 20 percent of models under \$75 certified, compared to rates as high as 78 percent for units between \$100 and \$199. This suggests that higher priced purifiers are much more likely to be energy efficient than lower priced units.

Intelligent controls also experienced a drastic price point increase, almost no units under \$100 were found to have intelligent controls, compared to a range of 45 to 60 percent of units over \$100. Such a large increase was not seen for mentions of COVID-19, which accounted for a small share of units within any price point, but accounted for the highest share of units between \$100 and \$199. Taken together, these results point to energy efficiency, automated controls, and pollution reduction as features much more likely to be found in higher-priced models, and consequently features that may be driving consumers who are purchasing more expensive purifiers.

Figure 14: Percentage of Units Sold per Price Bin by Marketing Terms

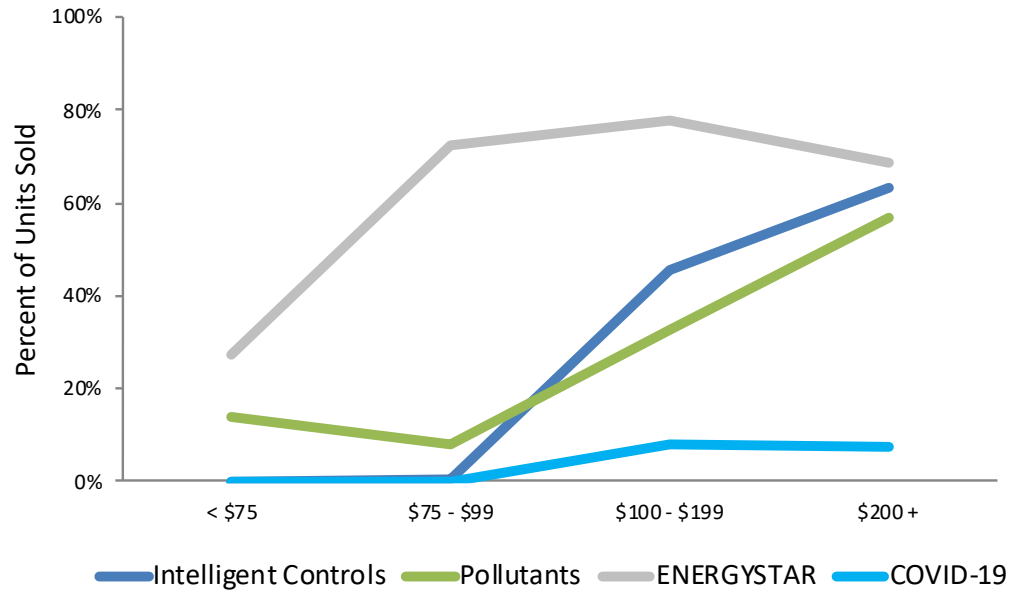


Figure 15 presents the percentage of units sold within each price bin that referenced “bacteria,” “viruses,” or “mold” in the product information found on the web that was joined onto NPD data. These three marketing terms are used most frequently for the units in the \$75-\$99 price range. This might suggest that the reduction of these particles may be most salient to customers who are in the market for moderately priced purifiers.

Figure 15: Percentage of Units Sold per Price Bin by Marketing Terms

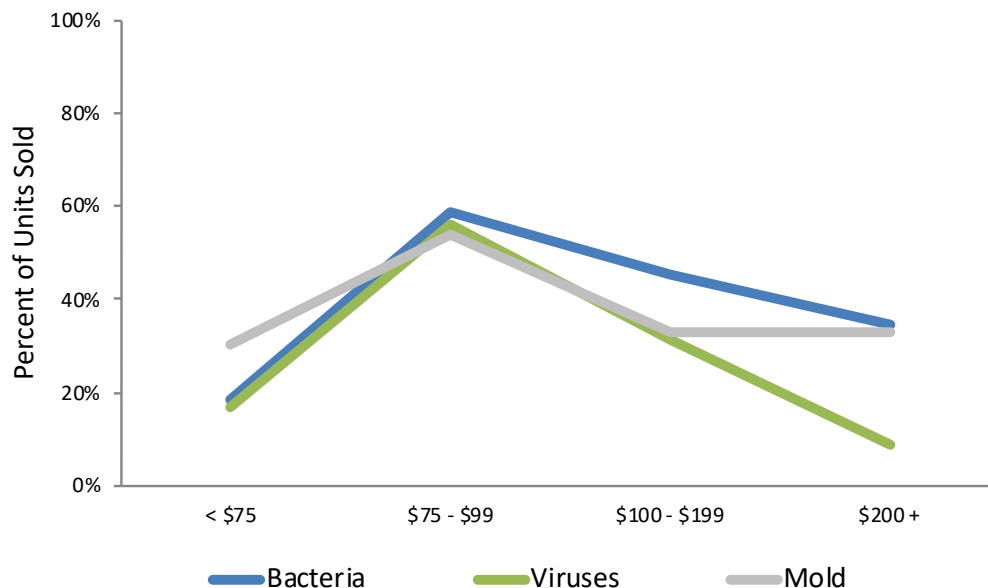


Figure 16, Figure 17, and Figure 18 present the percentage of units sold within each CADR bin by price. These figures show that the higher the CADR, the higher the price of the unit. The CADR information was gathered from data found on the web, where available, and joined onto NPD data. We could only search for CADRs for NPD records that had make and model information. Sixty-two percent of the units sold had a model number; however, we were unable to find CADR information for all model numbers. The units represented in Figure 16 through Figure 18 account for 41 percent of the total units sold between November 2020 and October 2021.

Figure 16 presents the percentage of units sold within each pollen CADR bin by price. While all the units with a pollen CADR rating of less than 50 cost less than \$75, a majority of the units with a CADR of 150 or higher cost more than \$200. This would imply that the amount of clean air delivered is the primary feature associated with price. A similar trend was seen for both dust and smoke CADRs as well, as shown in Figure 17 and Figure 18.

Figure 16: Percentage of Units Sold Within Pollen CADR Bins by Price

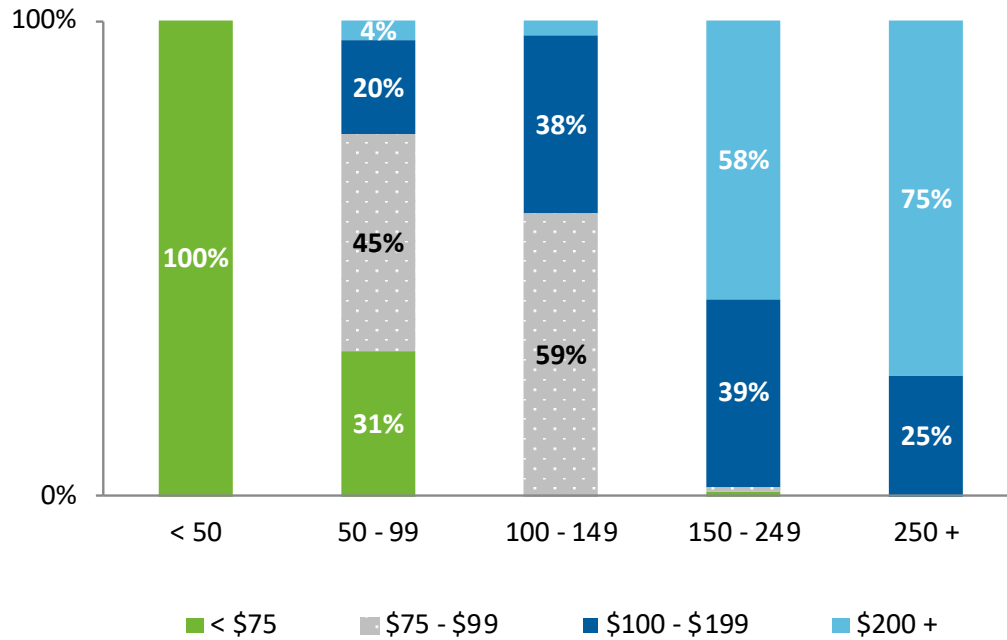


Figure 17: Percentage of Units Sold Within Dust CADR Bins by Price

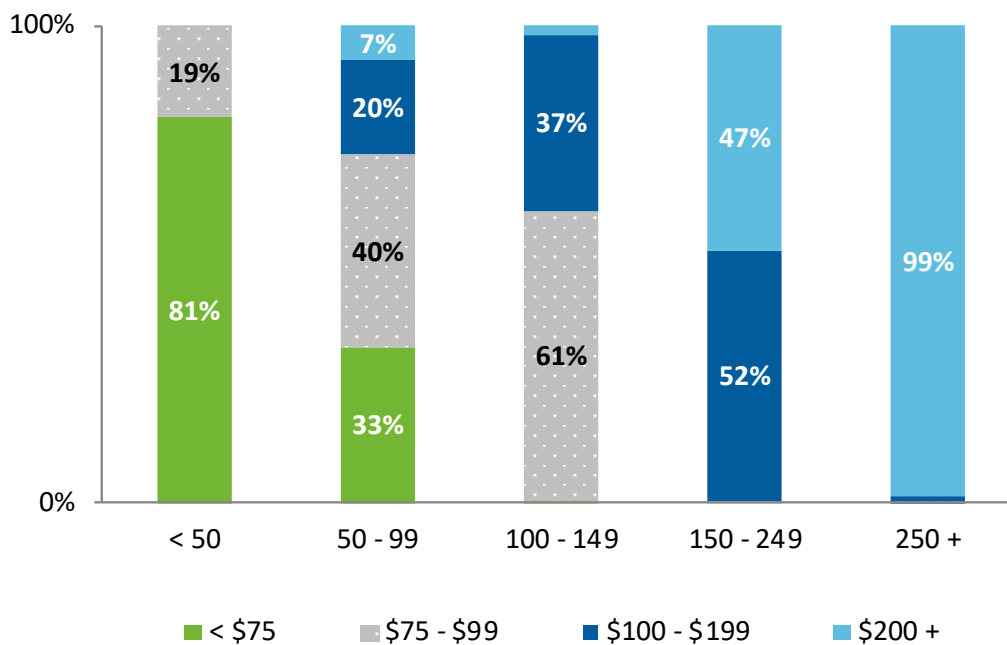
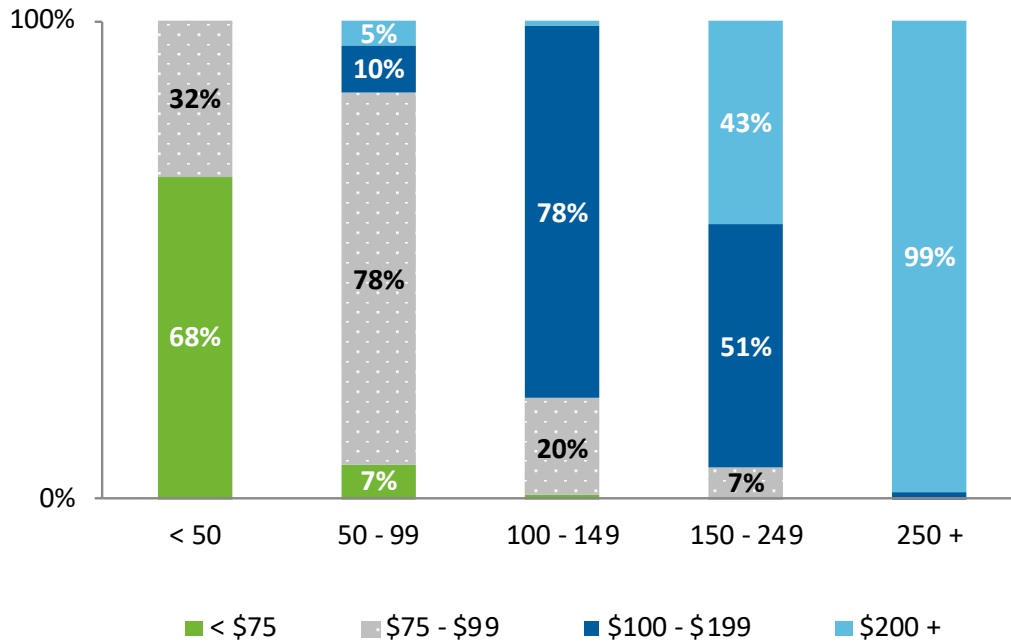


Figure 18: Percentage of Units Sold Within Tobacco Smoke CADR Bins by Price



Characteristics by Style

This next section presents the air purifier features found for different styles.

Table 11: Characteristics of Air Purifiers Sold Between November 2020 and October 2021 – NPD Data

Feature	Segment	Percentage of Units				
		Floor Model (N=3,045,977)	Tower (N=1,884,900)	Desktop (N=1,294,006)	Ash Tray/ Plug-In (N=202,159)	Personal (N=54,422)
Price	Average	\$169	\$147	\$112	\$19	\$31
	< \$75	5%	13%	57%	100%	100%
	\$75 - \$99	23%	35%	23%	0%	0%
	\$100 - \$199	40%	25%	6%	0%	0%
	\$200 +	32%	27%	15%	0%	0%
Season	Winter	23%	26%	28%	30%	35%
	Spring	24%	25%	23%	35%	20%
	Summer	29%	25%	23%	15%	20%
	Fall	25%	25%	25%	19%	25%
Ionizer	No Ionizer	88%	59%	88%	7%	77%
	Ionizer	12%	41%	12%	93%	23%

Table 12: Characteristics of Air Purifiers Sold Between November 2020 and October 2021 – Web Data

Feature ²⁵	Segment	Percentage of Units*				
		Floor Model (N=1,634,987)	Tower (N=1,476,436)	Desktop (N=844,907)	Ash Tray/ Plug-In (N=790)	Personal (N=41,785)
Certifications	ENERGY STAR	82%	79%	19%	0%	0%
	CARB	100%	100%	100%	100%	100%
Fan / Controls	Variable Speed Fan	100%	100%	100%	0%	71%
	Intelligent Controls	40%	38%	21%	0%	0%
Marketing Terms	Allergies	99%	100%	98%	0%	83%
	Odor	93%	93%	89%	100%	100%
	Pet Dander	80%	83%	96%	100%	83%
	Smoke	76%	82%	64%	0%	100%
	Bacteria	19%	71%	30%	0%	88%
	Mold	20%	66%	23%	0%	12%
	Pollutants	30%	40%	27%	0%	0%
	Viruses	7%	58%	11%	0%	88%
	COVID-19	11%	0%	1%	0%	0%
Marketed Room Size (Square Feet)	<i>Average</i>	<i>384 sq ft</i>	<i>229 sq ft</i>	<i>234 sq ft</i>	<i>324 sq ft</i>	<i>52 sq ft</i>
	< 150 Sq Ft	2%	4%	29%	0%	83%
	150 - 199 Sq Ft	22%	43%	22%	0%	0%
	200 - 299 Sq Ft	4%	10%	6%	0%	0%
	300 - 349 Sq Ft	17%	9%	0%	100%	0%
	350 - 399 Sq Ft	22%	16%	0%	0%	0%
	400 - 499 Sq Ft	11%	0%	0%	0%	0%
	500 + Sq Ft	22%	0%	20%	0%	0%
	Not Specified	1%	18%	23%	0%	17%

*Features in this table were merged by model number. Sixty-two percent of the units sold had a model number; the 38 percent without a model number are not represented in this table.

To better visualize the style data, figures created using the data from the tables above are presented below.

Figure 19 presents the percentage of units sold within each style by price bins segmented by Evergreen.

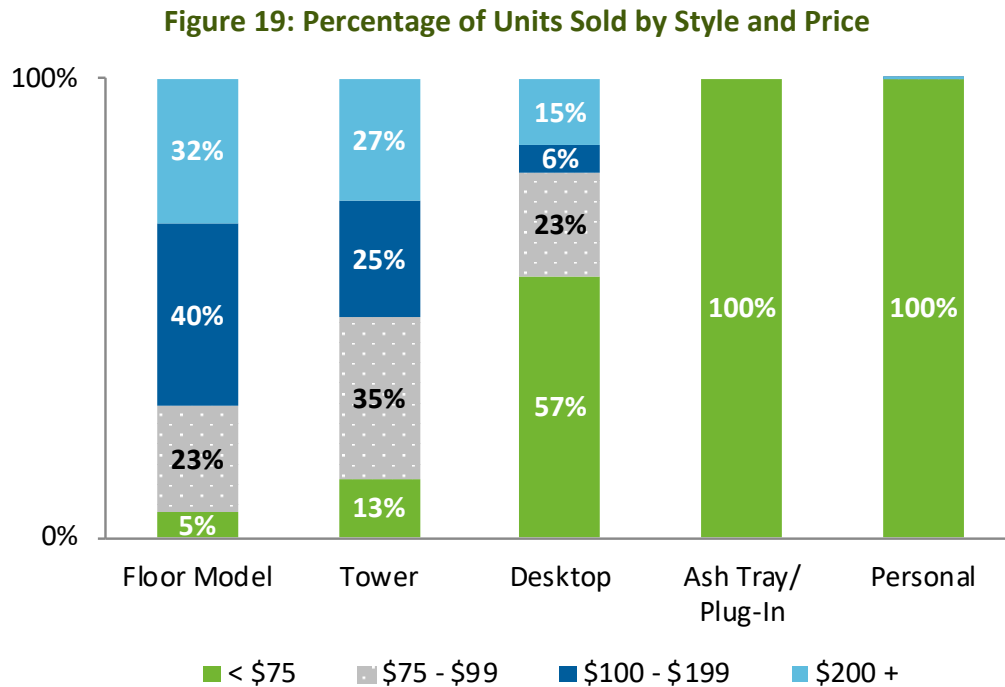


Figure 20 presents the percentage of units sold within each style that referenced “ENERGY STAR,” “intelligent controls,” “smoke,” or “pollutants” in the product information found on the web that was joined onto NPD data. Figure 21 presents the percentage of units sold within each style that referenced “mold,” “COVID-19,” “bacteria,” or “viruses” in the product information found on the web.

²⁵ None of the “Features” presented in this table originated from data provided by NPD.

Figure 20: Percentage of Units Sold by Style and Marketing Term

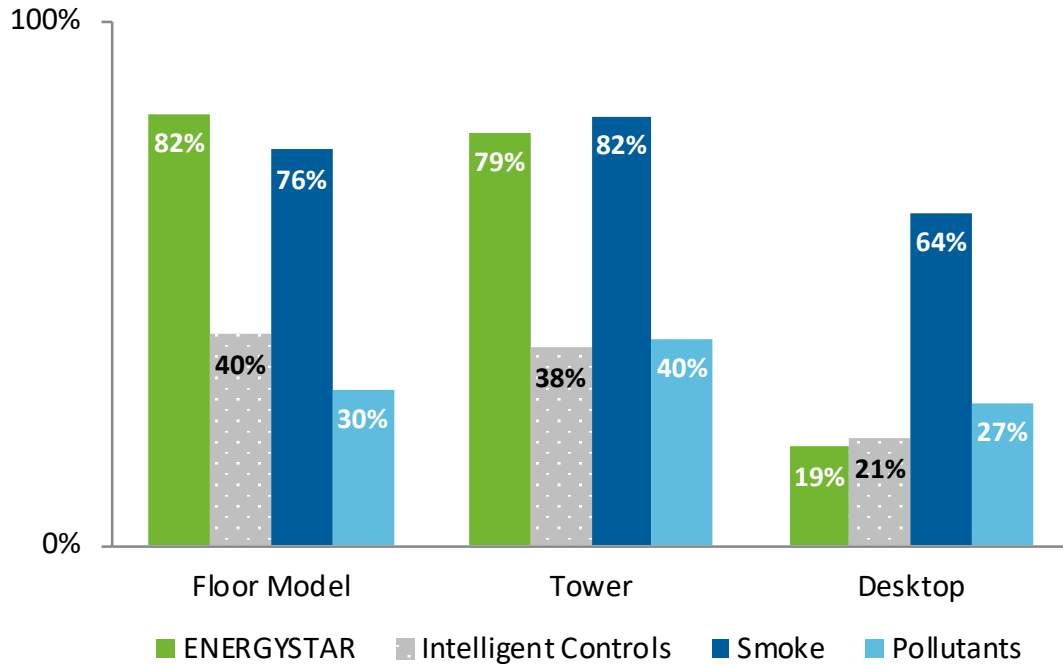


Figure 21: Percentage of Units Sold by Style and Marketing Term

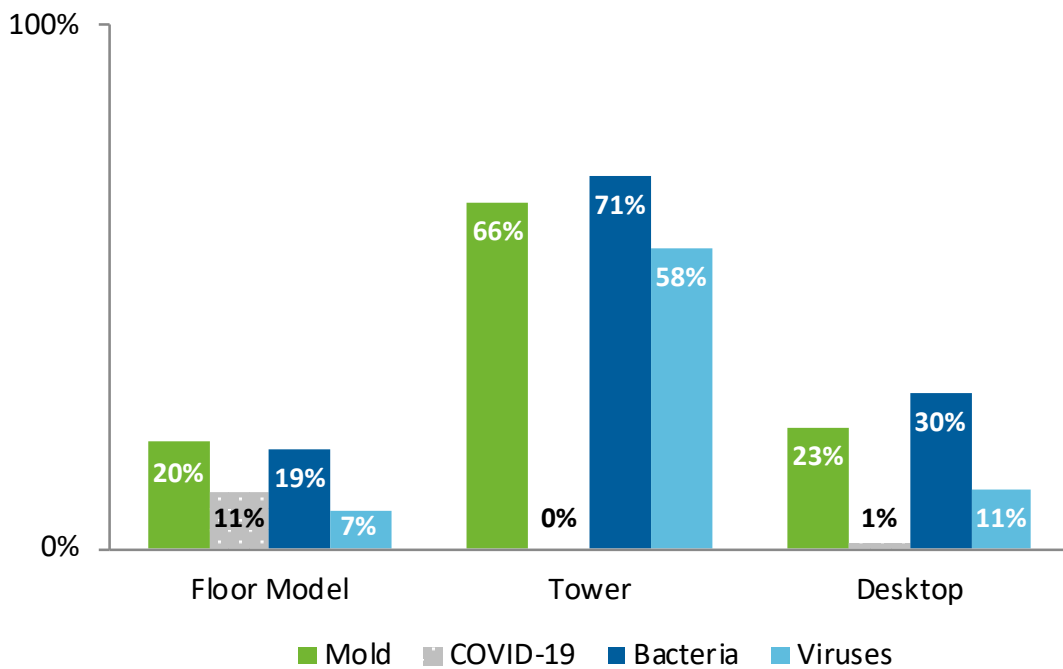


Figure 22, Figure 23, and Figure 24 present the percentage of units sold within each style by CADR bin. These figures show that the floor models had the highest CADRs, followed by towers and then desktop models. The CADR information was gathered from data found on the web, where available, and joined onto NPD data. We could only search for CADRs for NPD records that had make and model information. Sixty-two percent of the units sold had a model number; however, we were unable to find CADR information for all model numbers. The units represented in Figure 22 through Figure 24 account for 41 percent of the total units sold between November 2020 and October 2021.

Figure 22: Percentage of Units Sold by Style and Pollen CADR

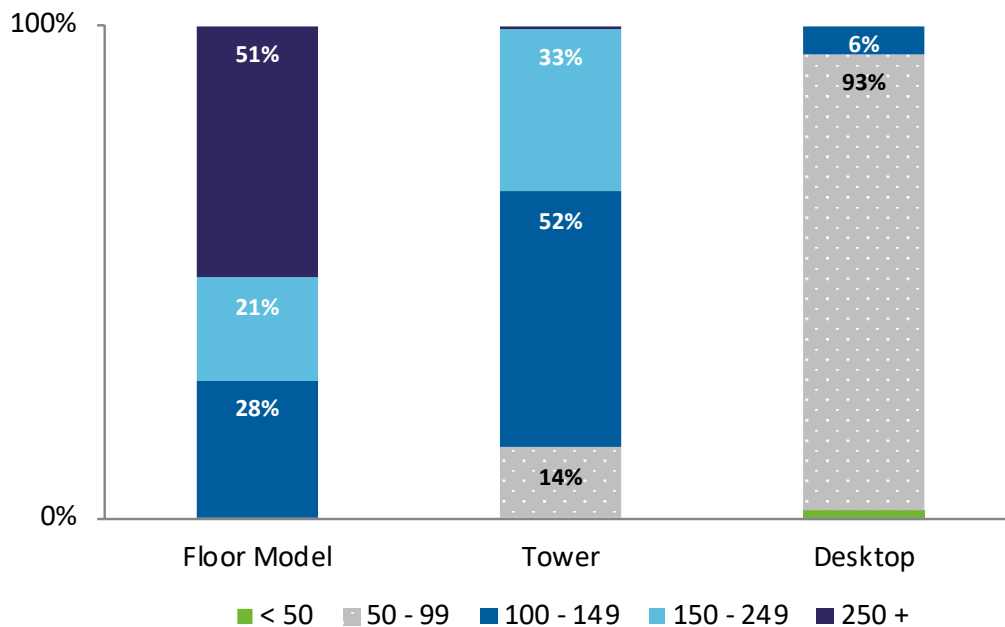


Figure 23: Percentage of Units Sold by Style and Dust CADR

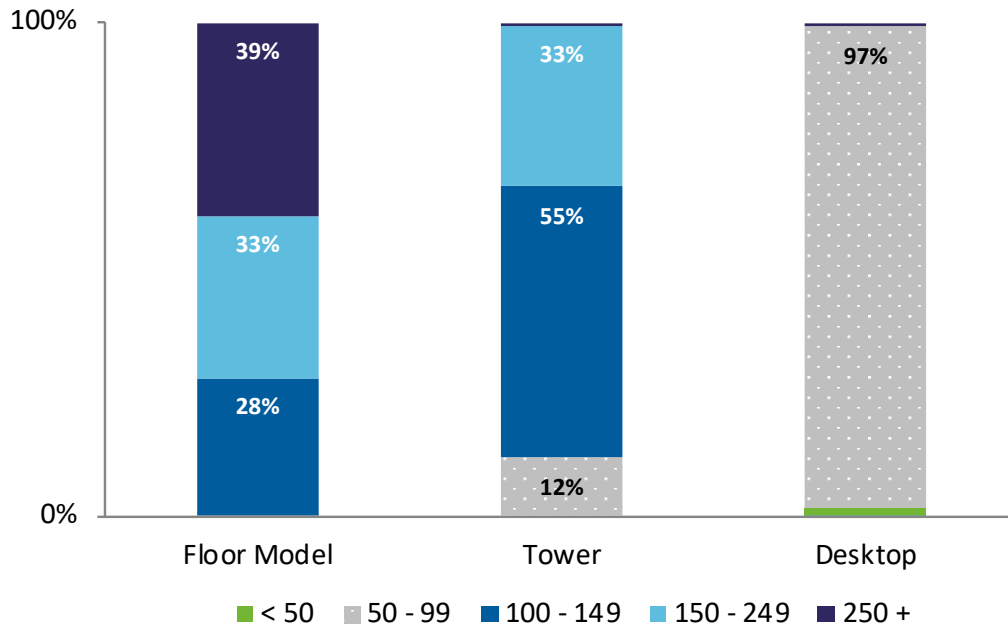
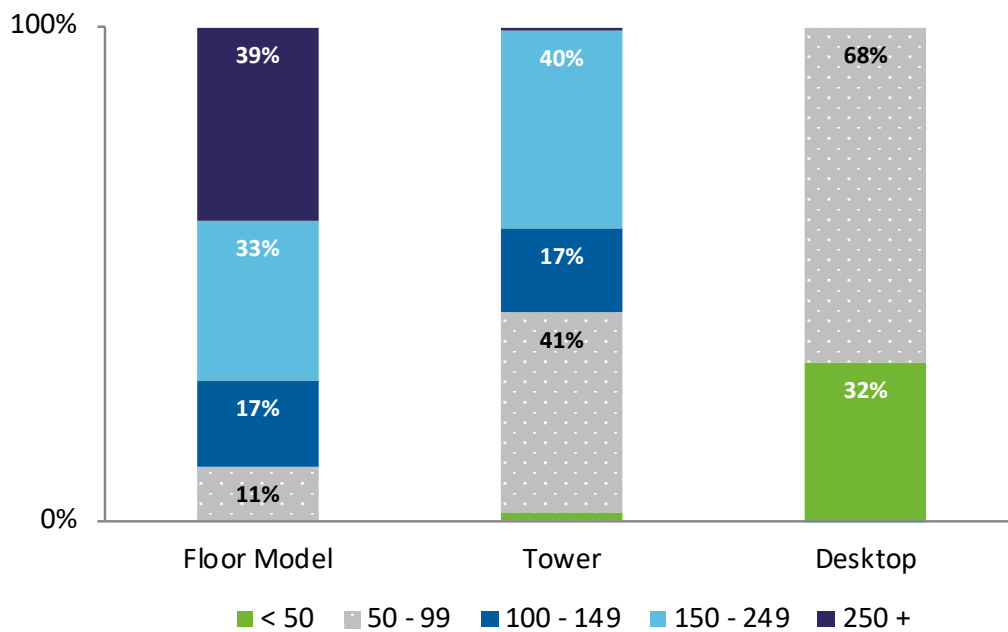


Figure 24: Percentage of Units Sold by Style and Tobacco Smoke CADR



Appendix A: Top 25 Models

The lists for the top 25-selling models in California and across the United States were very similar, with 23 models overlapping. Table 13 displays the top models and the units sold by geography.²⁶

Table 13: Top 25 Selling Models in California and the United States*

Brand	Model	California			United States		
		Rank	Net Units Sold	Net \$ Sold (in Millions)	Rank	Net Units Sold	Net \$ Sold (in Millions)
Germ Guardian	AC4825	1	57,863	\$4.8	1	353,903	\$29.1
Coway	AP1512HH	2	43,350	\$8.0	2	245,581	\$45.5
Levoit	LVH132	4	31,802	\$2.7	3	185,632	\$15.8
Levoit	VITAL100	5	27,557	\$3.2	4	160,084	\$18.7
Winix	WAC5500	7	25,750	\$4.0	5	149,228	\$23.1
Levoit	LVH128	8	24,935	\$1.1	6	148,519	\$6.8
Blueair	211PLUS	9	23,551	\$6.3	7	136,565	\$36.4
HoMedics	APT45	3	37,624	\$6.2	8	125,187	\$20.9
Honeywell	HPA300	10	20,522	\$4.5	9	124,176	\$27.5
Bionaire	BAP536UV	30	6,401	\$0.7	10	97,727	\$9.9
Dyson	30515801	6	26,279	\$9.0	11	95,658	\$33.4
Levoit	CORE200S	11	15,718	\$1.3	12	95,437	\$7.9
Blueair	411AUTO	13	13,450	\$1.7	13	75,956	\$9.7
Blueair	411	16	12,834	\$1.3	14	73,976	\$7.6
Dyson	31010501	12	13,771	\$5.9	15	73,279	\$31.0
Dyson	AM11	18	11,994	\$4.1	16	71,523	\$24.5
Germ Guardian	AC4300	17	12,292	\$1.2	17	70,090	\$6.8

²⁶ Rank is based on number of units sold between October 2020 to October 2021.

Brand	Model	California			United States		
		Rank	Net Units Sold	Net \$ Sold (in Millions)	Rank	Net Units Sold	Net \$ Sold (in Millions)
Honeywell	HPA200	21	8,463	\$1.5	18	60,169	\$10.5
Dyson	30557101	14	13,262	\$5.8	19	55,842	\$25.4
Germ Guardian	AC4100	20	8,774	\$0.5	20	48,892	\$2.9
HoMedics	APT20	23	7,663	\$0.9	21	45,487	\$5.0
Bissell	2780A	27	6,861	\$0.6	22	43,417	\$3.5
Dyson	24431401	22	8,072	\$4.0	23	42,089	\$20.8
Honeywell	HPA100	29	6,613	\$0.8	24	41,060	\$4.7
Pure Enrichment	PEAIRDSK	25	7,186	\$0.4	25	40,889	\$2.3
Winix	AM80	15	13,076	\$1.8	55	21,671	\$3.0
Coway	AP1518R	19	9,729	\$1.5	30	37,706	\$6.1
Pure Enrichment	PEPERSAP	24	7,304	\$0.2	28	39,587	\$1.3
All Others	with model #		249,652	\$50.4		1,616,857	\$312.4
All Others	without model #		427,118	\$49.2		2,707,016	\$282.9

*Data used for the top 25 model selection spanned from October 2020 to October 2021.

Table 14 displays the CADR ratings for the top models sold.

Table 14: CADR Ratings for the Top 25 Selling Models in California and the United States²⁷

Brand	Model	CA Rank	US Rank	CADR Dust	CADR Tobacco Smoke	CADR Pollen	CADR Room Size	Notes
Germ Guardian	AC4825	1	1	118	99	125	153	
Coway	AP1512HH	2	2	248	234	232	362	*Not on list
Levoit	LVH132	4	3					*Not on list
Levoit	VITAL100	5	4					*Not on list
Winix	WAC5500	7	5	216	235	251	350	
Levoit	LVH128	8	6					*Not on list
Blueair	211PLUS	9	7	350	350	350	540	
HoMedics	APT45	3	8	215	221	224	343	*Not on list
Honeywell	HPA300	10	9	320	300	300	465	
Bonaire	BAP536UV	30	10	137	154	149	239	
Dyson	30515801	6	11					*Not on list
Levoit	CORE200S	11	12	109	98	104	152	*Not on list
Blueair	411AUTO	13	13	120	105	120	161	
Blueair	411	16	14	120	105	120	161	
Dyson	31010501	12	15					*Not on list
Dyson	AM11	18	16					*Not on list
Germ Guardian	AC4300	17	17	107	99	116	153	
Honeywell	HPA200	21	18	190	200	180	310	
Dyson	30557101	14	19					*Not on list
Germ Guardian	AC4100	20	20	56	45	60	70	
HoMedics	APT20	23	21	93	110.4	91.9	171	*Not on list
Bissell	2780A	27	22	69	57	67	88	

²⁷ This segmentation into top 25 percent and all other models was done by Evergreen on the basis of NPD data. NPD did not segment the data in this manner. CADR information was not provided by NPD.

Brand	Model	CA Rank	US Rank	CADR Dust	CADR Tobacco Smoke	CADR Pollen	CADR Room Size	Notes
Dyson	24431401	22	23					*Not on list
Honeywell	HPA100	29	24	106	100	100	155	
Pure Enrichment	PEAIRDSK	25	25					*Not on list
Winix	AM80	15	55	240	233	230	360	
Coway	AP1518R	19	30	247	241	298	374	*Not on list
Pure Enrichment	PEPERSAP	24	28					*Not on list

*Not on AHAM's list of companies licensed to participate in the portable electric room air cleaners certification program